

FILIPPO MONTI

Los Angeles, CA, USA | <https://fil-monti.github.io>

Emails: fmonti@ucla.edu (work) | monti.filippo97@gmail.com (personal)

Research Interests: *Bayesian statistics, computational statistics, nonparametric methods, phylogenetics.*

Education

University of California, Los Angeles (UCLA), Los Angeles, USA

Ph.D. Candidate in Biostatistics

2022 – Present

Advisor: Prof. Marc A. Suchard

Bocconi University, Milan, Italy

M.Sc. in Economic and Social Sciences (ESS), *110/110 cum laude*

2019 – 2022

Thesis: *Bayesian nonparametric methods in Biostatistics*

Supervisor: Prof. Igor Prünster

Bocconi University, Milan, Italy

B.Sc. in Economic and Social Sciences (CLES), *110/110 cum laude*

2016 – 2019

Awards and Fellowships

Graduate Division Fellowship (funded), UCLA

2022 – 2026

Graduate Summer Research Mentorship (funded), UCLA

Summer 2023

Graduate Merit Award (full tuition waiver), Bocconi University

2019 – 2021

Unicredit Foundation Grant (full tuition waiver), BSE Summer School

Summer 2021

Publications and Working Papers

Frumence, E., Klitting, R., Serres, K., Shao, Y., **Monti, F.**, Vincent, M., Gill, M. S., Suchard, M. A., Lemey, P., de Lamballerie, X., Jaffar-Bandjee, M., and Dellicour, S. (2026). *Unravelling the epidemiological and dispersal dynamics of the 2024–2025 chikungunya virus epidemic on Réunion island*. In press, *Proceedings of the National Academy of Sciences*.

Monti, F., Ji, X., Shao, Y., Suchard, M. A. (2026+). *Nonparametric Modeling of Continuous-Time Markov Chains with Scalable Exact and Approximate Gradients*. R&R at JSSR-b.

Ji, X., Redelings, B., Su, S., Bao, H., Deng, W.-M., **Monti, F.**, Hong, S. L., Baele, G., Lemey, P., and Suchard, M. A. (2025). *Detecting Evolutionary Change-Points with Branch-Specific Substitution Models and Shrinkage Priors*. Minor Revision at Systematic Biology.

Monti, F., Faria, N. R., Ji, X., Lemey, P., Kraemer, M., and Suchard, M. A. (2026+). *Non-Linear Drivers of Population Dynamics: a Nonparametric Coalescent Approach*. Under review.

Monti, F., Holbrook, A., and Suchard, M. A. *Stable Matrix Parametrizations*

and Structured Adjoints for Ornstein–Uhlenbeck Processes. Under review.

Monti, F. and Suchard, M. A. *Markov Modulation via Reward Augmentation.* Manuscript in preparation.

Invited Talks and Conference Presentations

SysBio 2025 – Annual Systems Biology of Infectious Diseases Meeting, Rockville, MD, USA. Invited talk: *Flexibly Modeling Viral Substitution and Coalescent Processes with Covariates.*

BISP 2025 – Bayesian Inference in Stochastic Processes, Milan, Italy. Poster: *Nonparametric Modeling of Continuous-Time Markov Chains.*

ISBA World Meeting 2024, Venice, Italy. Poster: *Nonparametric Modeling of Continuous-Time Markov Chains.*

Teaching

Applied Bayesian Statistics School in Como (Italy)

Lecturer (with Prof. Suchard)

Bayesian phylogenetics and infectious diseases Summer 2024

University of California, Los Angeles (UCLA)

Teaching Assistant

Biostat 250C: Multivariate Biostatistics (PhD Level) Jan 2026 – March 2026

Biostat 250B: Linear Models (PhD Level) Jan 2026 – March 2026

Biostat 250A: Linear Models (PhD Level) Sept 2025 – Dec 2025

Biostat 100A/B: Introduction to Biostatistics Sept 2022 – Sept 2023

Other Appointments

University of California, Los Angeles (UCLA), Los Angeles, USA

Graduate Research Assistant, Suchard Group Jul 2022 – Present

Bocconi University, Milan, Italy

Fellow, Bayesian Learning Laboratory (BayesLab) 2021 – 2022

Visiting Student, BIDS-IGIER 2020 – 2021

Research Intern, IGIER (Supervisor: Prof. Marinacci) Summer 2018

Harvard University, Boston, USA

Fellow, Social Economics Lab 2020 – 2021

Research Assistant (hosted at IGIER, Bocconi) Nov 2020 – Jul 2021

European Commission, Luxembourg

Intern, DG ECFIN, Division L3: Risk Management Jun 2019 – Aug 2019

Service and Leadership

ISBA Bayesian Nonparametrics (BPN 2025), Los Angeles, USA

Volunteer, organizational support

Jul 2025

Bocconi University, Milan, Italy

President, Economic Society for Bocconi Students

Sep 2020 – Sep 2021

Student Representative, Council of the Department of Finance

2017 – 2019

Student Representative, Program Committee

2017 – 2018

Other Activities

Agesci (Italian Scout Association), Forlì, Italy

2006 – 2018

Volunteer, disadvantaged children's educational activities, Baia Mare, Romania

Aug 2016

Volunteer, elderly shelter, Forlì, Italy

Sep 2015 – Apr 2016

Skills

Programming: R, Java, Shell scripting (proficient); Julia, Python (good)

Languages: Italian (native), English (C2), Spanish (B2)

Last updated: August 17, 2026